

WashEase Project System Documentation

Complete File-by-File Usage, Architecture & Database Guide

1. Root Infrastructure Scripts

File / Component	Location	Usage & Functionality
start-all.bat	Root Directory	One-click Windows batch script to launch both PHP Backend (port 8000) and Vite React Frontend (port 5173).

2. Database Architecture

File / Component	Location	Usage & Functionality
schema.sql	database/	Master SQL schema script creating 13 relational MySQL tables (users, customers, vendors, orders, order_items, invoices, payments, complaints, fraud_alerts, reward_levels, notifications, otp_verifications, admin_settings, commission_payments).

3. Backend API Core & Configurations

File / Component	Location	Usage & Functionality
index.php	api/	Front Controller router handling action dispatching, CORS headers, JSON parsing, session auth, and role protection middleware.
database.php	api/config/	Database connection class implementing the Singleton pattern with PDO for MySQL connection management.
smtp.php	api/config/	SMTP configuration file returning mail settings (smtp_host, smtp_port, smtp_user, smtp_pass, from_name).

4. Backend API Controllers (api/controllers/)

File / Component	Location	Usage & Functionality
AuthController.php	api/controllers/	Handles registration, 2-step OTP email verification, Bcrypt login, logout, and password resets.
CustomerController.php	api/controllers/	Handles customer profile management and laundry shop location search.
VendorController.php	api/controllers/	Handles shop settings, reward metrics, sales analytics reports, and commission payment submissions.
OrderController.php	api/controllers/	Handles order creation, state machine updates, weight billing, invoice creation, and payment confirmation.
AdminController.php	api/controllers/	Handles platform analytics, user approvals, commission rules, vendor blocking, fraud alerts, and commission payment approvals.
ComplaintController.php	api/controllers/	Handles support ticketing, ticket assignment, progress tracking, and resolution notifications.

5. OOP Domain Models (api/models/)

File / Component	Location	Usage & Functionality
------------------	----------	-----------------------

File / Component	Location	Usage & Functionality
Customer.php	api/models/	Customer domain model extending User (fullName, address, placeOrder, trackOrder).
Vendor.php	api/models/	Vendor domain model extending User (shopId, ownerName, manageOrders, updateStatus).
Admin.php	api/models/	Admin domain model extending User (adminLevel, manageUsers, approveShops).
Order.php	api/models/	Order domain model (orderId, customerId, shopId, weight, specialNote, status, createdAt).
LaundryShop.php	api/models/	Laundry shop domain model (shopId, shopName, location, openTime, closeTime, servicesOffered).
Invoice.php	api/models/	Invoice domain model (invoiceId, orderId, invoiceNumber, totalAmount, laundryCharges, serviceCharges, taxes).
Complaint.php	api/models/	Support complaint domain model (complaintId, customerId, orderId, category, description, status, adminResponse).
CommissionLevel.php	api/models/	Vendor reward tier domain model (levelId, shopId, level, orderThreshold, commissionRate).
CommissionPayment.php	api/models/	Commission payment submission domain model (paymentId, vendorId, amount, transactionRef, status).
Payment.php	api/models/	Customer payment settlement domain model (paymentId, orderId, invoiceId, amountPaid, paymentStatus).
FraudAlert.php	api/models/	Discrepancy fraud alert domain model (alertId, orderId, customerAmount, vendorAmount, difference).
Notification.php	api/models/	System notification alert domain model (notificationId, userId, type, message, isRead, sentAt).

6. Backend Services (api/services/)

File / Component	Location	Usage & Functionality
MailService.php	api/services/	Sends 6-digit OTP verification emails via PHPMailer over SMTP.
PdfService.php	api/services/	Uses TCPDF library to dynamically generate downloadable PDF receipts (INV-XXXXXX) and sales reports.

7. Frontend Views & Components (frontend/src/pages/)

File / Component	Location	Usage & Functionality
Login.jsx	frontend/src/pages/	User login page with role redirection and password reset triggers.
Register.jsx	frontend/src/pages/	Registration page supporting Customer & Vendor signups with shop setup, GPS detection, and 2-step OTP verification popup.
CustomerDashboard.jsx	frontend/src/pages/	Customer workspace view featuring shop search, active order tracking timeline, booking modal, and payment confirmation.
VendorDashboard.jsx	frontend/src/pages/	Vendor workspace view featuring order queue management, clothes weight billing modal, shop profile settings, reward tier status, and commission payment submissions.

File / Component	Location	Usage & Functionality
Complaints.jsx	frontend/src/pages/	Customer/Vendor page for tracking and lodging support tickets.

8. Frontend Layout & CSS Design System

File / Component	Location	Usage & Functionality
App.jsx	frontend/src/	Main React root component providing top navbar, auth state, notification popover, and role routing.
main.jsx	frontend/src/	React DOM entry point mounting App.jsx into HTML element #root.
index.css	frontend/src/	Master Glassmorphism CSS design system defining dark/purple color palette, animations, glass cards, CSS Grid layouts, and status badges.
package.json	frontend/	Node.js package manifest listing frontend dependencies (React, Vite, Lucide-react) and scripts.
vite.config.js	frontend/	Vite development server configuration file.